Grader Output

C1M3_Programming

91 / 100 points earned

26 / 27 autograded cells passed

Graded Cells

Cell 3 (cell-fe8732a902953481)

Passed | 2.35 / 2.35 points
View feedback

Cell 5 (cell-d8e7049cf0e293d0)

Passed | 2.35 / 2.35 points
View feedback

Cell 7 (cell-356519bae70ac76e)

Passed | 0 / 0 points
View feedback

Cell 9 (cell-90f4f79771800969)

Passed | 3.53 / 3.53 points
View feedback

Cell 12 (cell-c8b4d2926f201f49)

Passed | 4.70 / 4.70 points
View feedback

Cell 14 (cell-4e3a9c0c167da7f6)

Passed | 2.35 / 2.35 points
View feedback

Cell 16 (cell-1733649e39b1b461)

Passed | 3.53 / 3.53 points
View feedback

Cell 18 (cell-663c0988ae5e335b)

Passed | 2.35 / 2.35 points
View feedback

Cell 21 (cell-32d100ed080e97c1)

Passed | 5.88 / 5.88 points
View feedback

Cell 23 (cell-a0c170c3f4748bbb)

Passed | 2.35 / 2.35 points
View feedback

Cell 25 (cell-51e047920bae4379)

Passed | 2.35 / 2.35 points
View feedback

Cell 27 (cell-6fb642d972bcc72a)

Passed | 3.53 / 3.53 points
View feedback

Cell 29 (cell-6302551bbcb36621)

Passed | 3.53 / 3.53 points
View feedback

Cell 31 (cell-2f8a5fb451559a03)

Passed | 2.35 / 2.35 points
View feedback

Cell 33 (cell-31a7e1ef597e0dd3)

Passed | 2.35 / 2.35 points
View feedback

Cell 35 (cell-d6dc58c8ff9fa1ed)

Passed | 5.88 / 5.88 points
View feedback

Cell 37 (cell-257a26302ebddbba)

Passed | 2.35 / 2.35 points
View feedback

Cell 39 (cell-c562bfd33da49eea)

Passed | 2.35 / 2.35 points
View feedback

Cell 41 (cell-3fe317a0b4c0c09f)

Passed | 2.35 / 2.35 points
View feedback

Cell 43 (cell-0ec58b70bed70ab5)

Passed | 3.53 / 3.53 points
View feedback

Cell 45 (cell-4d915569b4c8b04e)

Passed | 3.53 / 3.53 points
View feedback

Cell 47 (cell-37b031ea5fb0fb49)

Passed | 2.35 / 2.35 points
View feedback

Cell 49 (cell-f60c37ee3a0438ae)

Passed | 2.35 / 2.35 points
View feedback

Cell 51 (cell-fecfaa62cbad93fd)

Passed | 9.41 / 9.41 points
View feedback

Cell 53 (cell-ba8ef6e02f89c36f)

Try again | 0.00 / 9.41 points
View feedback

Cell 55 (cell-5a5b018640dc3734)

Passed | 5.88 / 5.88 points
View feedback

Cell 57 (cell-0be25422b796451f)

Passed | 7.05 / 7.05 points
View feedback

Module 3: Programming Assignment¶

In [1]:
```
library(testthat)
library(ggplot2)
```

Problem 1: Rate of wildfires¶

Imagine that, over the past 50 years Colorado has experienced an average of $\lambda_0 = 150$ large wildfires per year. For the next 10 years you will record the number of large fires in Colorado and then fit a Poisson/gamma model to these data. Let the rate of large fires in this future period, λ, have prior $\lambda \sim \Gamma(a, b)$.

Select a and b so that the prior is "uninformative" with prior variance around 100 and gives equal probability on the hypotheses

$$H_0 : \lambda \leq 150, \quad H_1 : \lambda > 150.$$

Note that, for $\lambda \sim \Gamma(a, b)$, $Var(\lambda) = \frac{a}{b^2}$, but that the median of λ does not have a nice closed form formula; so, you will need to do a grid search to find a and/or b.

1 (a) [2 points] Establish a dense grid of possible values for b using the function `seq()`. Your grid of possible values should contain `m=1000` evenly spaced values between 0.1 and 2. Store your grid of possible b values in a variable named `b_grid`.

In [2]: Student's answer (Top)
```
m = 1000
b_grid = seq(0.1, 2, length.out=m)
#b_grid
# your code here
#.NotYetImplemented()
```

In [3]: Grade cell: `cell-fe8732a902953481` Score: 2.6 / 2.6 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

1 (b) [2 points] Using `b_grid`, establish a vector of associated a values and call it `a_grid`. Note that we want the variance of the prior distribution on λ to be about 100 (use 100), and we know that for $\lambda \sim \Gamma(a, b)$, $Var(\lambda) = \frac{a}{b^2}$.

In [4]: Student's answer (Top)
```
a_grid = 100*b_grid^2
# your code here
#.NotYetImplemented()
#a_grid / b_grid^2
```

In [5]: Grade cell: `cell-d8e7049cf0e293d0` Score: 2.6 / 2.6 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

1 (c) [3 points] For each pair of components in `a_grid` and `b_grid`, calculate the median value using `qgamma()`. Use `a_grid` and `b_grid` as the shape and rate parameters (i.e. what we are calling a and b). Store these median values in a vector named `median_values`.

In [6]: Student's answer (Top)
```
median_values = rep(0, m)

for (i in 1:length(a_grid)) {
    median_values[i] <- qgamma(0.5, a_grid[i], b_grid[i])
}

#median_values
# your code here
#.NotYetImplemented()
```

In [7]: Grade cell: `cell-356519bae70ac76e` Score: 0.0 / 0.0 (Top)
```
Hidden Tests Redacted
```
```
Total possible points in this cell was 0.
```

1 (d) [3 points] Plot `median_values` vs. `b_grid` to visually estimate the value for b that ensures $\lambda = 150$. Calculate the exact value of `b_grid` minimizes the absolute distance between the median value established using `qgamma` and 150. Store this value in a variable named `b`. Using `b` calculate a and store it in a variable named `a`.

In [8]: Student's answer (Top)
```
plot(b_grid, median_values)
index <- which.min(abs(median_values - 150))
b = b_grid[index]
a = 100*b^2
# your code here
#.NotYetImplemented()
```

In [9]: Grade cell: `cell-90f4f79771800969` Score: 3.9 / 3.9 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

1 (e) [4 points] Ten years have gone by and the vector `fire` below represents the number of large fires in Colorado each year for these ten years. Calculate the posterior distribution using these data and the prior $\lambda \sim \Gamma(a, b)$ with the `a` and `b` computed above in 1 (d) for 200 evenly spaced λ values between 140 and 185 (i.e., use `seq(140,185,length.out=200)`). Store your grid of lambda values in a vector labeled `l_grid` and the values for the posterior in a vector labeled, `post`.

HINT: There is a conjugate relationship here to take advantage of!

In [10]:
```
fire = c(145, 142, 158, 143, 185, 164, 141, 148, 151, 150)
```

In [11]: Student's answer (Top)
```
l_grid = seq(140, 185, length.out=200)
#l_grid
#var(fire)
#a/b^2
post = dgamma(l_grid, a + sum(fire), b + length(fire))
#mean(rgamma(10000, a + sum(fire), b + length(fire)))
#(a + sum(fire)) / (b + length(fire))
# your code here
#.NotYetImplemented()
```

In [12]: Grade cell: `cell-c8b4d2926f201f49` Score: 5.19 / 5.19 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

1 (f) [2 points] Plot the posterior distribution computed in 1 (e) using ggplot. Save the plot to the variable `p`

In [13]: Student's answer (Top)
```
df = data.frame(lambda=l_grid, post=post)
p = ggplot(df, aes(lambda, post)) + geom_line()
p

# your code here
#.NotYetImplemented()
```

In [14]: Grade cell: `cell-4e3a9c0c167da7f6` Score: 2.6 / 2.6 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

1 (g) [3 points] Compute a 90% credible interval and store it in a vector labeled, `cred_int`.

In [15]: Student's answer (Top)
```
cred_int = c(qgamma((1-0.9)/2, a + sum(fire), b + length(fire)), qgamma(1-(1-0.9)/2, a + sum(fire), b + length(fire)))
cred_int
# your code here
#.NotYetImplemented()
```

In [16]: Grade cell: `cell-1733649e39b1b461` Score: 3.9 / 3.9 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

1 (h) [2 points] [Multiple Choice] Do you have reason to believe that the rate of wildfires is different across these ten years when compared to past data?
Select the best response to this question and storing the number associated with the response in a variable labelled `choice`.

1. No. The mean of the posterior is very close to 150, and the credible interval covers $\lambda = 150$. Given this, we do not have reason to believe that the rate has changed.
2. No. When we look at the values in `fire` we see that they are, for the most part, very close to 150. Given this, we do not have reason to believe that the rate has changed.
3. Yes. The mean of the posterior is clearly above 150 as observed in the plot in part 1 (f) and the shape of the posterior indicates we are very certain that the true mean is above 150.

For example, if you believe the best response is #3, code: `choice=3` in the code block below.

In [17]: Student's answer (Top)
```
choice = 1

# your code here
#.NotYetImplemented()
```

In [18]: Grade cell: `cell-663c0988ae5e335b` Score: 2.6 / 2.6 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

Problem 2: Normal conjugate example¶

Suppose Company A is considering using recycled cobalt in the smartphone batteries that it produces. Denote the true mean battery life (run time on a full charge) as μ. Company A would like to know if there is evidence to strongly suggest that μ exceeds or falls short of 30 hours (30 hours is the true mean battery life of the battery currently in use). Researchers' prior beliefs about μ can be represented by a normal distribution, centered at $\theta = 30$, with variance $\tau^2 = 4$. That is: $\mu \sim N(30, 4)$.

Suppose the researchers at company A test $n = 5$ randomly selected phones with the recycled cobalt. Denote the battery life of phones in a possible sample as $X_1, \ldots, X_n$. Assume that the standard deviation of battery life times is $\sigma = 1$) For this particular sample, the data $x_1, \ldots, x_n$ were measured as:

In [19]:
```
x = c(28.3, 28.1, 29.1, 30, 30.1)
```

2 (a) [5 points] Compute the posterior distribution mean and variance of $\mu \mid \mathbf{x}$ and store them in variables labeled `bat_mean` and `bat_var`, respectively.

In [20]: Student's answer (Top)
```
n = length(x)
sigma2 = 1
xbar = mean(x)
bat_mean = (n*xbar/sigma2 + 30/4) / (n/sigma2 + 1/4)
bat_var = 1 / (n/sigma2 + 1/4)

#mean(x)
#var(x)
bat_mean
bat_var

# your code here
#.NotYetImplemented()
```

In [21]: Grade cell: `cell-32d100ed080e97c1` Score: 6.49 / 6.49 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

2 (b) [2 points] [Multiple Choice] What would be a reasonable point estimate for μ given the data?
Select the best response to this question and storing the number associated with the response in a variable labelled `choice`.

1. $\widehat{\mu} \approx 0.19$
2. $\widehat{\mu} \approx 29.2$
3. $\widehat{\mu} \approx 14.6$

For example, if you believe the best response is #3, code: `choice=3` in the code block below.

In [22]: Student's answer (Top)
```
choice = 2

# your code here
#.NotYetImplemented()
```

In [23]: Grade cell: `cell-a0c170c3f4748bbb` Score: 2.6 / 2.6 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

2 (c) [2 points] Calculate the prior distribution for a grid 300 evenly spaced values of μ between 20 and 40 (i.e., use `seq(20,40,length.out=300)`. Store the prior distribution in a vector labeled, `bat_prior`.

In [24]: Student's answer (Top)
```
bat_grid = seq(20, 40, length.out=300)
bat_prior = dnorm(bat_grid, 30, 2)

# your code here
#.NotYetImplemented()
```

In [25]: Grade cell: `cell-51e047920bae4379` Score: 2.6 / 2.6 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

2 (d) [3 points] Calculate the posterior distribution for the same grid of evenly spaced values of μ between 20 and 40. Store the posterior distribution in a vector labeled, `bat_post`. Using ggplot, plot the prior distribution and posterior distribution (on the same plot). Store the plot in a variable labeled, `bat_p`.

In [26]: Student's answer (Top)
```
bat_post = dnorm(bat_grid, bat_mean, sqrt(bat_var))
df = data.frame(bat_life=bat_grid, prior=bat_prior, posterior=bat_post)
bat_p = ggplot(df) + geom_line(aes(bat_life, prior, col='prior')) +
                     geom_line(aes(bat_life, posterior, col='posterior')) +
                     theme_bw()
bat_p

# your code here
#.NotYetImplemented()
```

In [27]: Grade cell: `cell-6fb642d972bcc72a` Score: 3.9 / 3.9 (Top)
```
Hidden Tests Redacted
```
```
Congratulations! All test cases in this cell passed.
```

2 (e) [3 points] [Multiple Choice] What do you notice about the preceding plot?

Select the best response to this question and storing the number associated with the response in a variable labelled `choice`.

1. The prior reflects a high degree of uncertainty in the value of μ. The posterior reflects less uncertainty.
2. The posterior reflects a high degree of uncertainty in the value of μ. The prior reflects less uncertainty.
3. The posterior mean is shifted above the prior because the data suggest the true mean is greater than the researchers' hypothesis that μ is centered at 30.

For example, if you believe the best response is #3, code: `choice=3` in the code block below.

In [28]: Student's answer (Top)